

Explaining Quantum Superposition and Entanglement

Quantum Lego Pinball & Goldberg Machine

June 4th, 2025

Team Quant



Quantum Computer

A next-generation computer that processes information using the principles of quantum mechanics

	Classic Computer	Quantum Computer
Unit of Information	bit (0 or 1)	qubit (Superposition of 0 and 1)
Computational Method	Sequential computation	Parallel computation

**Quantum
Superposition**

A quantum system can exist in multiple states simultaneously, each with a certain probability

If Particle A is in a superposition of 0 and 1, $A = \alpha|0\rangle + \beta|1\rangle$

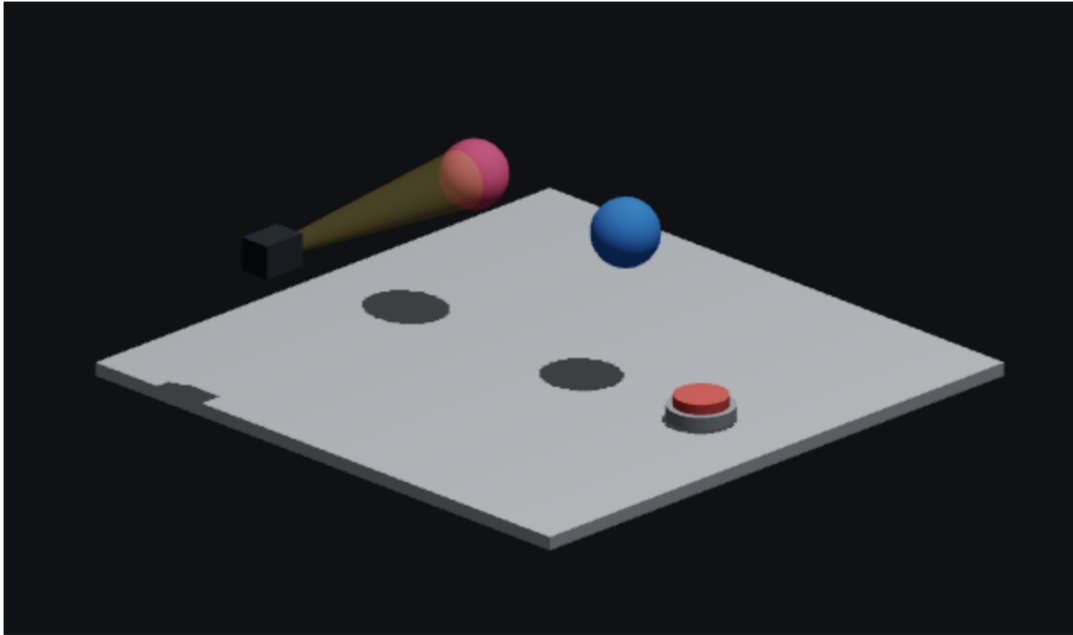
$|\alpha|^2$: $|0\rangle$ Probability of observing each state

$|\beta|^2$: $|1\rangle$ Probability of observing each state

**Quantum
Entanglement**

A phenomenon in which two or more quantum particles are strongly correlated, such that measuring one particle determines the corresponding state of the other

Visualizing the Principles of Quantum Computing

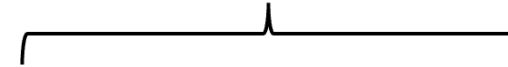


(<https://qubit.donghwi.dev/basic>)

Basic Principles of Quantum Computing

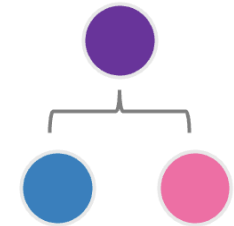
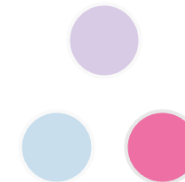
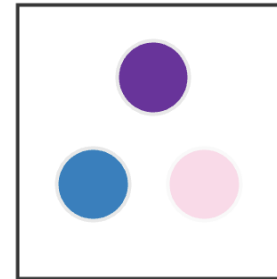


Core Concepts of Quantum Mechanics

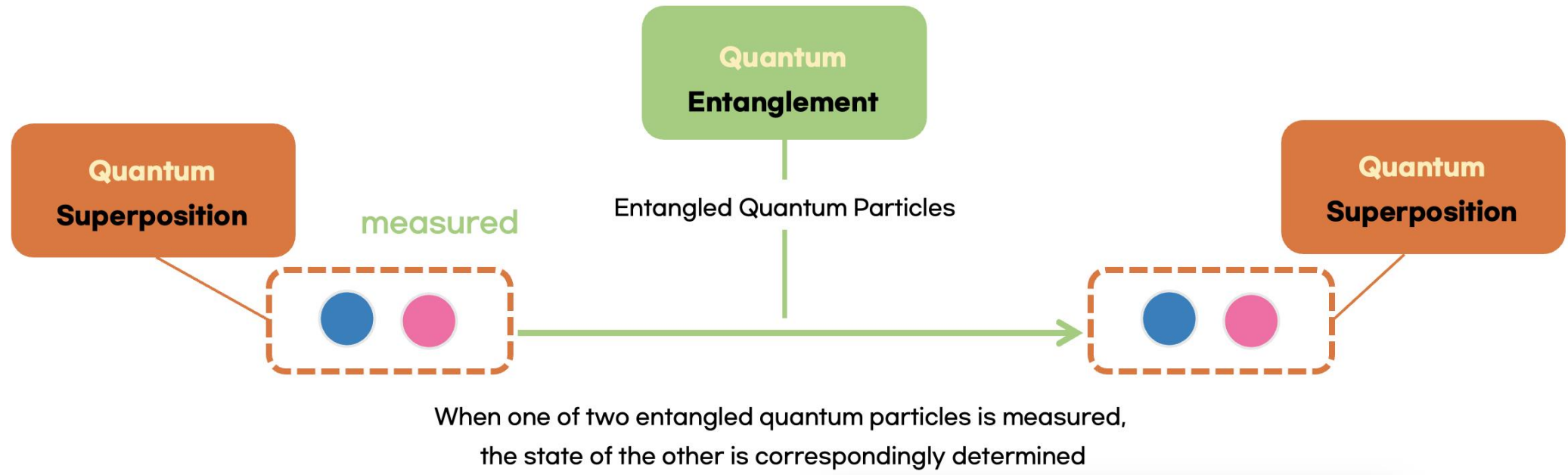


Quantum
Entanglement

Quantum
Superposition



How Can We Visualize It?



Superposition Pinball Machine

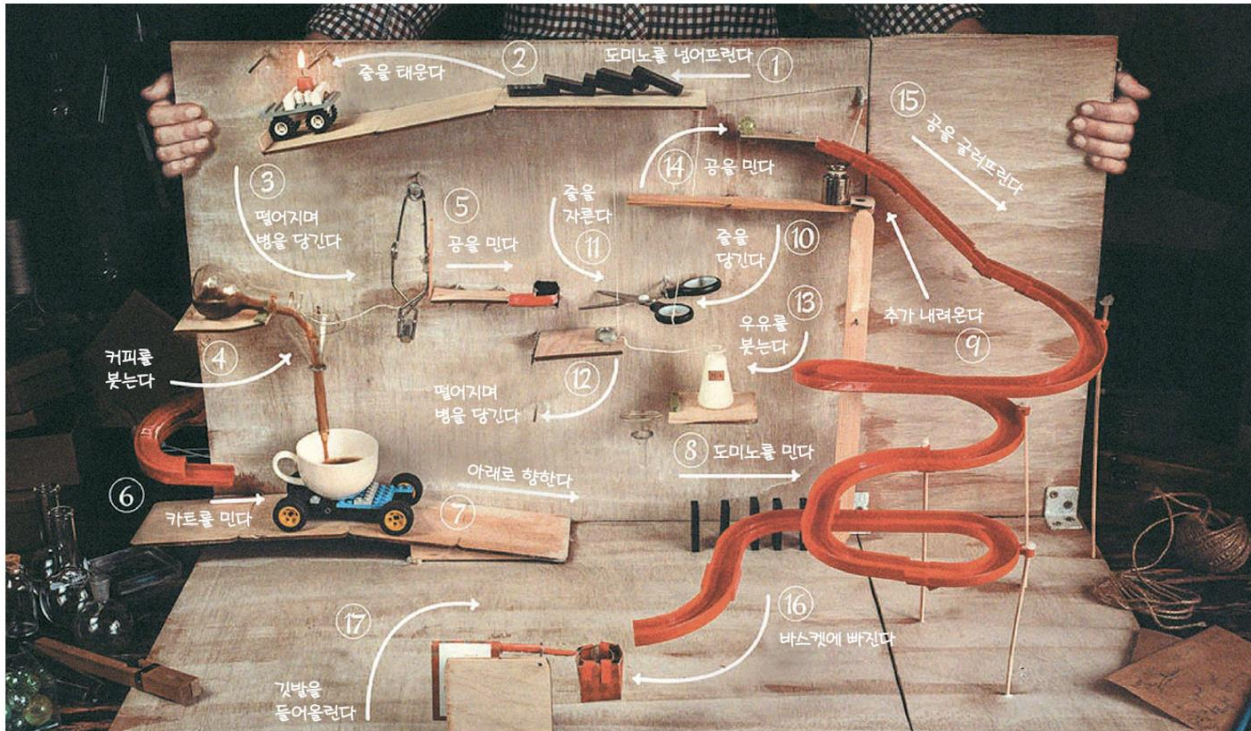


Entanglement Observation Rube Goldberg Machine

Why a Rube Goldberg Machine?

Goldberg Machine

Named after American cartoonist Rube Goldberg, a Rube Goldberg machine is a chain-reaction device designed to perform a simple task through a series of indirect and elaborate steps.



(<https://www.joongang.co.kr/article/18648599>)

- By making the **process** leading to an outcome directly visible, a Rube Goldberg machine can help visualize the fundamental quantum computing concepts of superposition and entanglement.
- By prompting questions such as “*How does it work?*” and “*What will the final outcome be?*”, the machine **stimulates viewers’ curiosity and imagination**, helping them better understand and engage with the concepts of superposition and entanglement.
- Our goal is to present the principles of quantum computing in an intuitive and accessible way, even **for those without a technical background**.

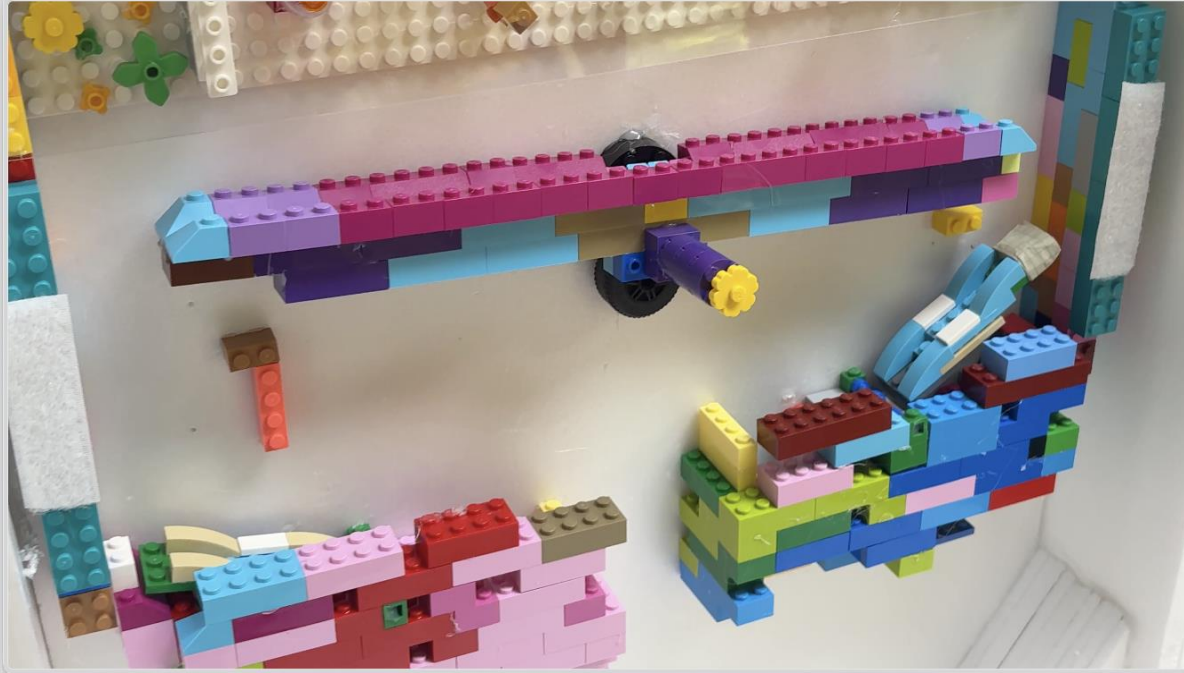
1. Pinball Machine (front)



1. Pinball Machine (front)

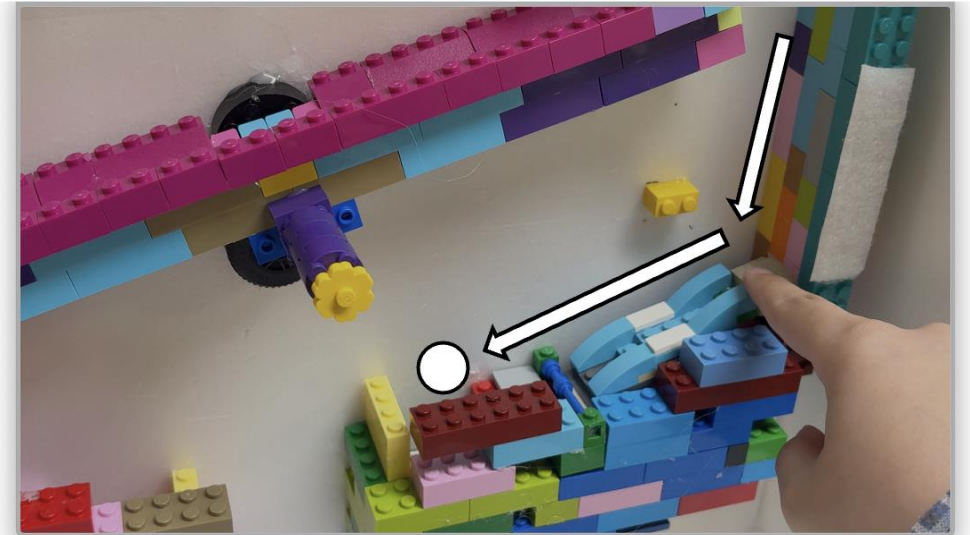
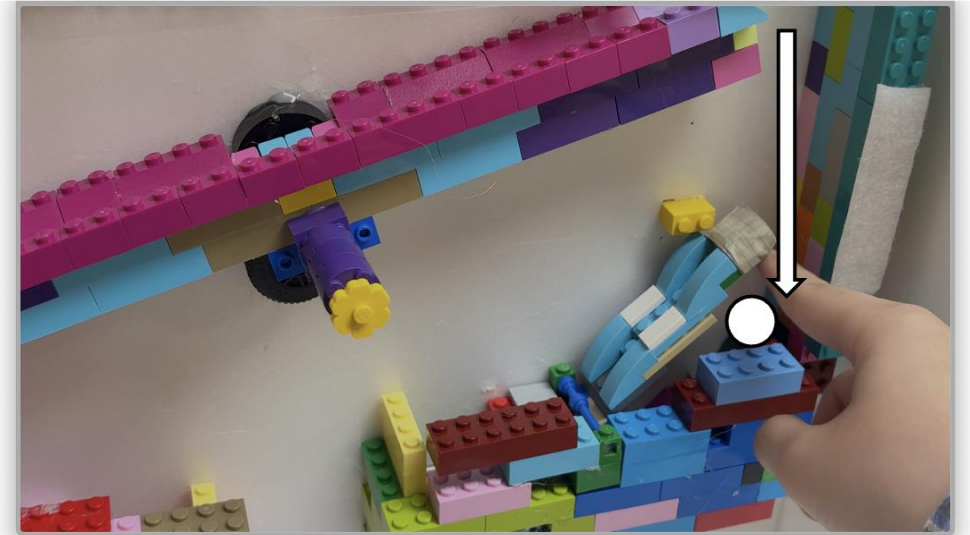
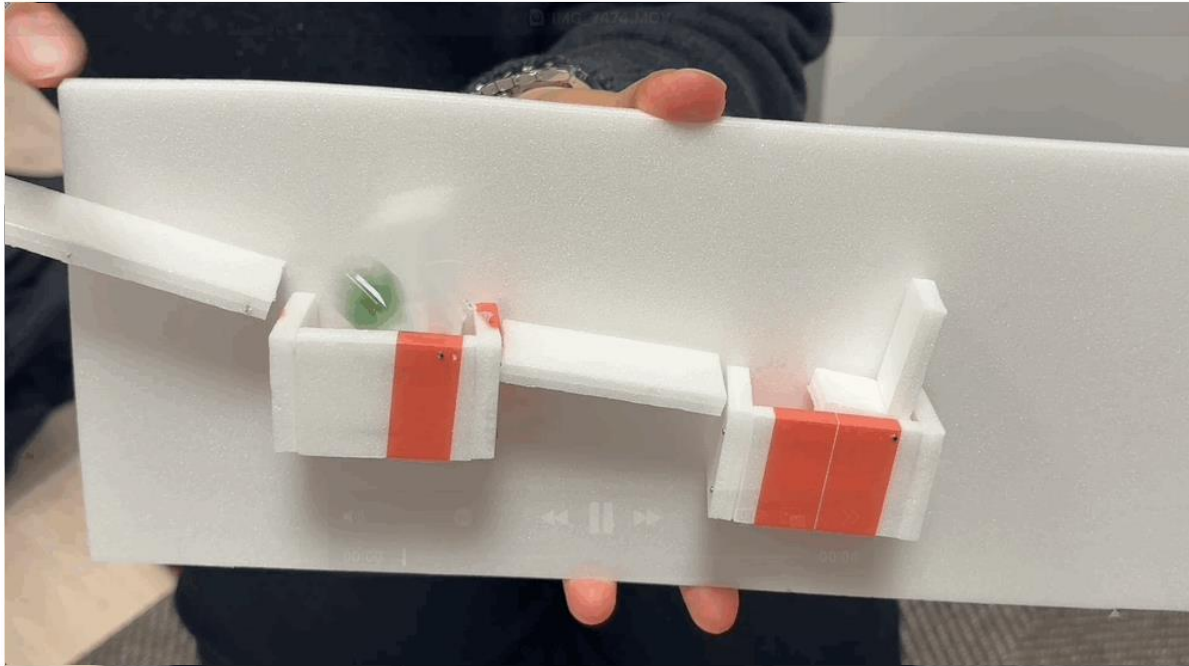


1. Pinball Machine (front)



1. Pinball Machine (front)

Where the state of each quantum particle is stored (representing superposition)



2. Goldberg Machine (back)



2. Goldberg Machine (back)

A component linked to the front mechanism, where each quantum particle's state is stored



2. Goldberg Machine (back)

Measurement

=

the motion of the large ball through the Goldberg machine

